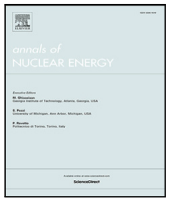
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Development of a diffusive simplified double P_N approximation to the neutron transport equation

A. Carreño [a], A. Vidal-Ferràndiz [b], D. Ginestar [b], G. Verdú [a,*]

[a] Instituto Universitario de Seguridad Industrial, Radiofísica y Medioambiental, Universitat Politècnica de València, Camino de Vera, s/n, 46022, Valencia, Spain
[b] Instituto Universitario de Matemática Multidisciplinar, Universitat Politècnica de València, Camino de Vera, s/n, 46022, Valencia, Spain

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ABSTRACT

The neutron transport equation models the distribution of the neutron power inside a reactor core. The complexity of integrating this equation implies the use of some angular approximations. Discrete ordinates and spherical harmonics equations (P_N equations) are commonly used to this purpose, but even when these approaches are used, the resulting set of equations requires high computational demands. An alternative is to develop simplified formulations of them, such as the simplified spherical harmonics equations (SP_N equations), which provide accurate results. The spherical harmonic equations are based on the regularity of solution in terms of the angular dependence. However, in some reactor configurations, the angular fluxes can be discontinuous. For that, the double spherical harmonics equations, where angular fluxes are expanded in terms of two sets of Legendre polynomials, were studied. In this work, a simplified treatment of them is developed that leads to a diffusive set of equations, the simplified double spherical harmonics equation (SDP_N equations). Numerical results show that the SDP_N approximation provides faster and more accurate results than the SP_N equations for some problems with strong variations of the angular neutron flux.

1. Introduction

Discrete ordinates or spherical harmonics are discretization methods which are commonly employed to approximate the neutron transport equation (Larsen and Keller, 1974; Lewis and Miller, 1984). However, when dealing with a realistic nuclear reactor core, these approximations demand extremely high computational resources that cannot be accessible to most of the nuclear analysis (Hamilton and Evans, 2015; Alexander et al., 2020). The neutron diffusion equation has been the most popular approximation across time because, despite some restrictions on its application, it provides accurate enough results at a reasonable computational cost. Nevertheless, better approximations of the neutron transport equation must be considered when novel fuel types and setups with complicated geometry are wanted to be analyzed (Nuclear Energy Agency (NEA), 2003). When a quadrature set is used to discretize the angular variable of the neutron flux, the discrete ordinates (S_N) equations are obtained (Lewis and Miller, 1984; Chen et al., 2017). Furthermore, the method of characteristics is used to approximate the neutron transport equation by discretizing the angular dependence using a quadrature set and integrating the multi-group characteristic form of the equations (Askew, 1972; Boyd et al., 2014). An alternative approximation, the P_N equations, is obtained by expanding the angular variable in terms of spherical harmonics (Fletcher, 1983; Capilla et al., 2020).

For a realistic system, all these approximations are composed of a large number of equations, which require an enormous computational cost to be solved (Jarrell et al., 2013). To reduce these high computational requirements, a simplified formulation of the P_N equations, defined as the simplified P_N equations (SP_N equations), was developed (Gelbard, 1960; McClarren, 2010). This approximation starts from the P_N equations in slab geometry, and the partial derivatives of the even moments are then replaced by gradient operators and the derivatives of the odd moments are replaced by divergence operators. The theoretical basis for the SP_N equations has been studied, for example, in McClarren (2010), Brantley and Larsen (2000) and Sanchez (2019), and this approximation is widely used due to its reasonable computational cost (Carreño et al., 2022; Albadr et al., 2023).

Since this kind of approximation relies on expanding the 1D-angular flux in terms of Legendre polynomials, it is assumed that this flux is a continuous function of the angular variable. Nevertheless, there are situations such as the presence of purely absorbing materials or sources, which give rise to discontinuous angular fluxes (Greenspan et al.,

* Corresponding author.
E-mail addresses: amcarsan@mat.upv.es (A. Carreño), anvifer2@upv.es (A. Vidal-Ferràndiz), dginesta@mat.upv.es (D. Ginestar), gverdu@iqn.upv.es (G. Verdú).

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1968). This motivated the introduction of the Double P_N approximation (Yvon, 1957) based on the introduction of two sets of polynomials that can improve the behavior of the neutron flux expansions in these situations. Recently, it has been suggested to use the Double P_N approximation (DP_N equations) to construct a Simplified DP_N approximation to deal with multidimensional problems (Nazari et al., 2023). One of the attractiveness of the SP_N approximation is that these equations have a diffusive behavior and can be easily implemented, making small modifications to the methods based in the diffusion approximation. In this way, here we present a derivation of the Simplified Double P_N approximation that has a 'diffusive form' and can be used as an alternative to the diffusive SP_N equations. We test its performance by solving different benchmark problems.

To study the accuracy and the computational performance of the SDP_N approximation, they are implemented in the open-source code `FEMFFUSION` (Vidal-Ferràndiz et al., 2020; Fontenla et al., 2024). This code is build on the top of the open-source libraries `deal.ii` (Bangerth et al., 2007), `PETSc` (Balay et al., 2015) and `SLEPc` (Hernandez et al., 2005). `FEMFFUSION` uses the continuous Galerkin finite element method to be able to deal with any type of geometry, structured or unstructured, and problem dimension.

The paper is structured as follows. Section 2 shows the process that leads to the simplified spherical harmonics equations from the one-dimensional neutron transport equation. Section 3 presents the one-dimensional double spherical harmonics equations used in this work to develop the simplified formulation of them. Section 4 tests the developed formulations for different benchmark problems. Finally, Section 5 summarizes the main results of this paper.

2. Simplified P_N equations

2.1. SP_N Equations

Our starting point is the steady-state multigroup neutron transport equation for the slab geometry (Bell and Glasstone, 1970),

$$\mu\frac{d}{dx}\psi_g(x,\mu)+\Sigma_{tg}(x)\psi_g(x,\mu)-\sum_{g'=1}^{G}\int_0^{2\pi}d\tilde{\varphi}\int_{-1}^{+1}\Sigma_{sgg'}\left(x,\mu_0\right)\psi_{g'}\left(x,\tilde{\mu}\right)\,\tilde{\mu}$$
$$=\frac{1}{2}\frac{\chi_g}{k_{\text{eff}}}\sum_{g'=1}^{G}\nu_{g'}\Sigma_{fg'}(x)\int_{-1}^{+1}\psi_{g'}\left(x,\tilde{\mu}\right)\,d\tilde{\mu},\qquad g=1,\ldots,G, \tag{1}$$

where $\psi_g(x,\mu)$ is the angular neutron flux of the gth energy group that depends on the position $x\in[x_L,x_R]$, where x_L and x_R are left and right boundaries of the nuclear system, and the direction cosine $\mu=\cos(\theta)$, where θ is the angle between the incident neutrons and the x axis. The variable $\tilde{\mu}$ represents the other cosine direction that is changed by scattering or by isotropic fission (including μ). Hence, $\mu_0=\cos(\theta_0)$, where θ_0 is the angle between the direction of the incident neutrons and the direction of the scattered neutrons. φ and $\tilde{\varphi}$ are azimuthal angles corresponding to the direction of the incident neutrons and the scattered neutrons, respectively. The variable g' indicates the energy group (including g) of the neutron flux after change by scattering or by fission. The coefficient k_{eff} is the ratio of neutron population that shows the state of criticality of the reactor. Magnitudes, Σ_{tg}, $\Sigma_{sgg'}$, Σ_{fg} are the total, the scattering and the fission macroscopic cross-sections, respectively. Finally, ν_g and χ_g are the mean number of fission neutrons and the fission spectrum.

It is assumed the following expansion using the Legendre polynomials $P_l(\mu)$ as

$$\Sigma_{sgg'}\left(x,\mu_0\right)=\sum_{l=0}^{\infty}\frac{2l+1}{4\pi}\Sigma^l_{sgg'}(x)P_l\left(\mu_0\right)\ . \tag{2}$$

Application of the addition theorem of Legendre polynomials (Bell and Glasstone, 1970; Stacey, 2007) leads to

$$P_l\left(\mu_0\right)=P_l(\mu)P_l\left(\tilde{\mu}\right)+2\sum_{m=1}^{l}\frac{(l-m)!}{(l+m)!}P_l^m(\mu)P_l\left(\tilde{\mu}\right)\cos(m(\varphi-\tilde{\varphi}))\ ,$$

where $P_l^m(\mu)$ are the associated Legendre polynomials, since,

$$\int_0^{2\pi}\cos(m(\varphi-\tilde{\varphi}))\,d\tilde{\varphi}=0,$$

we have that

$$\mu\frac{d}{dx}\psi_g(x,\mu)+\Sigma_{tg}(x)\psi_g(x,\mu)$$
$$=\sum_{g'=1}^{G}\sum_{l=0}^{\infty}\frac{2l+1}{2}\Sigma^l_{sgg'}(x)P_l(\mu)\int_{-1}^{+1}P_l\left(\tilde{\mu}\right)\psi_{g'}\left(x,\tilde{\mu}\right)\,d\tilde{\mu}$$
$$+\frac{1}{2}\frac{\chi_g}{k_{\text{eff}}}\sum_{g'=1}^{G}\nu_{g'}\Sigma_{fg'}(x)\int_{-1}^{+1}\psi_{g'}\left(x,\tilde{\mu}\right)\,d\tilde{\mu}\ . \tag{3}$$

We assume that the angular neutron flux also admits an expansion of the form

$$\psi_g(x,\mu)=\sum_{l=0}^{\infty}\phi^l_g(x)\left(\frac{2l+1}{2}\right)P_l(\mu)\ , \tag{4}$$

and thus, Eq. (3) can be rewritten as

$$\sum_{l=0}^{\infty}\frac{d}{dx}\phi^l_g(x)\left(\frac{2l+1}{2}\right)\mu P_l(\mu)+\Sigma_{t,g}(x)\sum_{l=0}^{\infty}\phi^l_g(x)\left(\frac{2l+1}{2}\right)P_l(\mu)$$
$$=\sum_{g'=1}^{G}\sum_{l=0}^{\infty}\left(\frac{2l+1}{2}\right)\Sigma^l_{sgg'}(x)P_l(\mu)\int_{-1}^{+1}P_l\left(\tilde{\mu}\right)\sum_{s=0}^{\infty}\left(\frac{2s+1}{2}\right)\phi^s_{g'}(x)P_s\left(\tilde{\mu}\right)\,d\tilde{\mu}$$
$$+\frac{\chi_g}{2k_{\text{eff}}}\sum_{g'=1}^{G}\nu_g\Sigma_{fg'}(x)\int_{-1}^{+1}\sum_{s=0}^{\infty}\left(\frac{2s+1}{2}\right)\phi^s_{g'}(x)P_s\left(\tilde{\mu}\right)\,d\tilde{\mu}\ . \tag{5}$$

Applying the orthogonality relation for the Legendre polynomials

$$\int_{-1}^{+1}P_m(\mu)P_n(\mu)\,d\mu=\frac{2}{2n+1}\delta_{m,n}, \tag{6}$$

where $\delta_{n,m}$ is the Kronecker delta, and the recurrence relation

$$\mu P_l(\mu)=\frac{1}{2l+1}\left((l+1)P_{l+1}(\mu)+lP_{l-1}(\mu)\right)\ , \tag{7}$$

it is possible to express Eq. (5) as

$$\sum_{l=0}^{\infty}\frac{d}{dx}\phi^l_g(x)\frac{1}{2}\left((l+1)P_{l+1}(\mu)+lP_{l-1}(\mu)\right)+\Sigma_{tg}(x)\sum_{l=0}^{\infty}\phi^l_g(x)\left(\frac{2l+1}{2}\right)P_l(\mu)$$
$$=\sum_{g'=1}^{G}\sum_{l=0}^{\infty}\left(\frac{2l+1}{2}\right)\Sigma^l_{sgg'}(x)P_l(\mu)\phi^l_g(x)+\frac{\chi_g}{2k_{\text{eff}}}\sum_{g'=1}^{G}\nu_g\Sigma_{fg'}(x)\phi^0_{g'}(x)\ . \tag{8}$$

Multiplying Eq. (8) by $P_n(\mu)$, integrating from -1 to $+1$, and applying the orthogonality relation (6) results in the P_N equations in slab geometry

$$\frac{n}{(2n+1)}\frac{d}{dx}\phi^{n-1}_g(x)+\frac{n+1}{(2n+1)}\frac{d}{dx}\phi^{n+1}_g(x)(x)+\Sigma_{tg}(x)\phi^n_g(x) \tag{9}$$
$$=\sum_{g'=1}^{\infty}\Sigma^n_{sgg'}(x)\phi^n_{g'}(x)+\frac{\chi_g}{k_{\text{eff}}}\sum_{g'=1}^{G}\nu_{g'}\Sigma_{fg'}(x)\phi^0_{g'}(x)\delta_{0,n}\ . \tag{10}$$

In this formulation the expansions are truncated in such a way that $n=0,1,\ldots,N$. Moreover, the scattering coefficients are considered to be zero for moments greater than N. The set of Eqs. (10) consists of $N+1$ equations with $N+2$ unknowns. Different possibilities are discussed for the closure of the P_N equations (Hauck and McClarren, 2010). Usually, it is solved by imposing the condition $\frac{d}{dx}\phi^{N+1}_g=0$.

The 1D P_N equations can be extended for multi-dimensional problems by using a simplified approach, the SP_N equations. They are obtained by replacing, in the P_N Eqs. (10), the partial derivative with respect to x of the odd moments by the divergence operator and the partial derivatives of the even moments, by the gradient operator (Gelbard, 1960; Brantley and Larsen, 2000; McClarren, 2010). This formulation leads to $(N+1)/2$ diffusion-like equations, by removing the odd moments terms (Hamilton and Evans, 2015; Vidal-Ferràndiz et al., 2019).

The SP_N equations can be expressed in matrix notation over energy groups as

$$-\nabla\cdot\left[\frac{n\,(\mathbf{\Sigma^{n-1}})^{-1}}{(2n+1)(2n-1)}\nabla\left((n-1)\phi^{n-2}+n\phi^{n}\right)+\frac{(n+1)(\mathbf{\Sigma^{n+1}})^{-1}}{(2n+1)(2n+3)}\nabla\left((n+1)\phi^{n}+(n+2)\phi^{n+2}\right)\right]+\mathbf{\Sigma^{n}}\phi^{n}=\frac{1}{k_{\text{eff}}}\mathbf{F}\phi^{n}\delta_{n0}\,,\qquad n=0,2,\ldots,N-1\,. \tag{11}$$

where,

$$\phi^{n}=\begin{pmatrix}\phi_1^n\\ \phi_2^n\\ \vdots\\ \phi_G^n\end{pmatrix},\quad \mathbf{\Sigma^{n}}=\begin{pmatrix}\Sigma_t^1-\Sigma_{s11}^n & -\Sigma_{s12}^n & \cdots & -\Sigma_{s1G}^n\\ -\Sigma_{s21}^n & \Sigma_{t2}-\Sigma_{s22}^n & \cdots & -\Sigma_{s2G}^n\\ \vdots & \vdots & \ddots & \vdots\\ -\Sigma_{sG1}^n & -\Sigma_{sG2}^n & \cdots & \Sigma_{tG}-\Sigma_{sGG}^n\end{pmatrix},$$

$$\mathbf{F}=\begin{pmatrix}\chi_1\nu_1\Sigma_{f1} & \chi_1\nu_2\Sigma_{f2} & \cdots & \chi_1\nu_G\Sigma_{fG}\\ \chi_2\nu_1\Sigma_{f1} & \chi_2\nu_2\Sigma_{f2} & \cdots & \chi_2\nu_G\Sigma_{fG}\\ \vdots & \vdots & \ddots & \vdots\\ \chi_G\nu_1\Sigma_{f1} & \chi_G\nu_2\Sigma_{f2} & \cdots & \chi^G\nu_G\Sigma_{fG}\end{pmatrix}. \tag{12}$$

Finally, the change of variables introduced in Hamilton and Evans (2015) is considered

$$U^m=(2m-1)\,\phi^{2m-2}+2m\,\phi^{2m},\qquad m=1,2,\ldots,M-1\,,$$
$$U^M=(2M-1)\,\phi^{2M-2}, \tag{13}$$

where the vector U^m encloses the diffusive pseudo-moments

$$U^m=(u_1^m,u_2^m,\ldots,u_G^m)^{\mathsf{T}}\,, \tag{14}$$

to obtain a system (after an appropriate reordering) of the form

$$-\nabla\cdot(\mathbb{D}\nabla U)+\mathbb{A}U=\frac{1}{\lambda}\mathbb{F}U\,. \tag{15}$$

In case of the SP_7 equations, these matrices are given by

$$U=\begin{pmatrix}U^1\\U^2\\U^3\\U^4\end{pmatrix},\qquad \mathbb{D}=\begin{pmatrix}\frac{1}{3}(\Sigma^1)^{-1} & 0 & 0 & 0\\ 0 & \frac{1}{7}(\Sigma^3)^{-1} & 0 & 0\\ 0 & 0 & \frac{1}{11}(\Sigma^5)^{-1} & 0\\ 0 & 0 & 0 & \frac{1}{15}(\Sigma^7)^{-1}\end{pmatrix},$$

$$\mathbb{A}_{ab}=\sum_{m=1}^{4}c_{ab}^{(m)}\mathbf{\Sigma}^m,\qquad \mathbb{F}_{ab}=c_{ab}^{(1)}\mathbf{F}, \tag{16}$$

and the coefficients matrices, $c^{(m)}$, are

$$c^{(1)}=\begin{pmatrix}1 & -\frac{2}{3} & \frac{8}{15} & -\frac{16}{35}\\ -\frac{2}{3} & \frac{4}{9} & -\frac{16}{45} & \frac{32}{105}\\ \frac{8}{15} & -\frac{16}{45} & \frac{64}{225} & -\frac{128}{525}\\ -\frac{16}{35} & \frac{32}{105} & -\frac{128}{525} & \frac{256}{1225}\end{pmatrix},\quad c^{(2)}=\begin{pmatrix}0 & 0 & 0 & 0\\ 0 & \frac{5}{9} & -\frac{4}{9} & \frac{8}{21}\\ 0 & -\frac{4}{9} & \frac{16}{45} & -\frac{32}{105}\\ 0 & \frac{8}{21} & -\frac{32}{105} & \frac{64}{245}\end{pmatrix},$$

$$c^{(3)}=\begin{pmatrix}0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0\\ 0 & 0 & \frac{9}{25} & -\frac{54}{175}\\ 0 & 0 & -\frac{54}{175} & \frac{324}{1225}\end{pmatrix},\quad c^{(4)}=\begin{pmatrix}0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0\\ 0 & 0 & 0 & \frac{13}{49}\end{pmatrix}. \tag{17}$$

SP_1 equations are obtained by setting $\phi^2=\phi^4=\phi^6=0$. The resulting system is similar to the standard multigroup neutron diffusion equation. SP_3 equations are got by substituting $\phi^4=\phi^6=0$, and, finally, SP_5 equations are obtained by replacing $\phi^6=0$.

2.2. Boundary conditions for SP_N equations

Vacuum boundary conditions for the one-dimensional problem (1) are imposed at $x=x_L$ and $x=x_R$ as

$$\psi_g(x_L,\mu_{\text{in}})=0=\psi_g(x_R,\mu_{\text{in}})\,, \tag{18}$$

where $\psi_g(x,\mu_{\text{in}})$ is the incident angular flux in the incoming direction.

For the P_N approximation, we use the Marshak boundary conditions for vacuum boundary conditions (18) that are expressed as Stacey (2007)

$$\int_{\mu_{in}}P_n(\mu)\sum_{n'=0}^{N}\frac{2n'+1}{2}\phi^{n'}(x_0)P_{n'}(\mu)\mathrm{d}\mu=0\,,\quad n=1,3,\ldots,N\,. \tag{19}$$

In the P_7 equations, at the left boundary $x=x_L$, the Marshak conditions are:

$$\begin{aligned}\frac{1}{2}\phi^0+\frac{5}{8}\phi^2-\frac{3}{16}\phi^4+\frac{13}{128}\phi^6&=-\phi^1\,,\\ -\frac{1}{8}\phi^0+\frac{5}{8}\phi^2+\frac{81}{128}\phi^4-\frac{13}{64}\phi^6&=-\phi^3\,.\\ \frac{1}{16}\phi^0-\frac{25}{128}\phi^2+\frac{81}{128}\phi^4+\frac{325}{512}\phi^6&=-\phi^5\,.\\ -\frac{5}{128}\phi^0+\frac{7}{64}\phi^2-\frac{105}{512}\phi^4+\frac{325}{512}\phi^6&=-\phi^7\,.\end{aligned} \tag{20}$$

Marshak boundary conditions in the simplified case are defined applying the same strategy that was applied in P_N Eqs. (10), that is

$$\pm\frac{d}{dx}\to\vec{n}\cdot\nabla, \tag{21}$$

where $\vec{n}$ is the normal vector at the boundary.

Using the previous considerations and the change of variable (13) yields the SP_N Marshak boundary conditions (Hamilton and Evans, 2015)

$$-\vec{n}\,\mathbb{D}\vec{\nabla}U=\mathbb{B}U\,. \tag{22}$$

For the SP_7 case, the matrix $\mathbb{B}$ is defined by the Kronecker product of $\mathbf{g}$ by the identity matrix of dimension $G\times G$ as

$$\mathbb{B}=\mathbf{g}\otimes\mathbf{I}_{(G\times G)}\,,\qquad \text{where}\qquad \mathbf{g}=\begin{pmatrix}\frac{1}{2} & -\frac{1}{8} & \frac{1}{16} & -\frac{5}{128}\\ -\frac{1}{8} & \frac{7}{24} & -\frac{41}{384} & \frac{1}{16}\\ \frac{1}{16} & -\frac{41}{384} & \frac{407}{1920} & -\frac{233}{2560}\\ -\frac{5}{128} & \frac{1}{16} & -\frac{233}{2560} & \frac{3023}{17920}\end{pmatrix}. \tag{23}$$

Moreover, one can consider in the problem (1) symmetry or, reflecting boundary conditions by forcing

$$\psi_g(x_L,\mu)=\psi_g(x_L,-\mu),\qquad \psi_g(x_R,\mu)=\psi_g(x_R,-\mu)\,. \tag{24}$$

Considering the expansion (4) and noticing that odd Legendre polynomials are different for $-\mu$ and μ, one can deduce that the reflecting conditions (24) for P_N equations implies

$$\phi^n(x_L)=0=\phi^n(x_R),\qquad n=1,3,\ldots,N\,. \tag{25}$$

For the SP_N Eqs. (15) reflecting boundary conditions are then set as Hamilton and Evans (2015)

$$\vec{\nabla}U^m=0,\qquad m=1,2,\ldots,(N+1)/2\,. \tag{26}$$

3. SDP_N equations

3.1. DP_N equations

In order to improve the convergence of the angular flux in the expansion of polynomials when this flux becomes discontinuous (Gelbard, 1960), two sets of polynomials are introduced,

$$P_l^+(\mu)=\begin{cases}P_l(2\mu-1), & \text{if } \mu\geq 0,\\ 0, & \text{if } \mu<0,\end{cases}\qquad P_l^-(\mu)=\begin{cases}0, & \text{if } \mu\geq 0,\\ P_l(2\mu+1), & \text{if } \mu<0,\end{cases}, \tag{27}$$

that allow to expand the flux independently in each half-space $0\leq\mu\leq 1$ and $-1\leq\mu<0$ (Stacey, 2007).

The polynomials (27) satisfy

$$\int_0^{+1} P_l^+(\mu)P_m^+(\mu)\,d\mu = \int_{-1}^{0} P_l^-(\mu)P_m^-(\mu)\,d\mu = \frac{\delta_{l,m}}{2l+1}\,. \tag{28}$$

To obtain the classical double P_N formulation, it is assumed that

$$\psi_g(x,\mu) = \sum_{l=0}^{\infty}(2l+1)\left(\phi_g^{l,+}(x)P_l^+(\mu) + \phi_g^{l,-}(x)P_{l,g}^-(\mu)\right).$$

However, as we want to obtain a diffusive-like set of equations in terms of ϕ_g^n, $n = 0,1,\ldots,N$, we directly use the expansion (4). Multiplying the Eqs. (8) by $P_n^+(\mu)$ and integrating μ between 0 and 1, we obtain,

$$\begin{aligned}&\sum_{l=0}^{\infty}\frac{d}{dx}\phi_g^l(x)\frac{1}{2}\left((l+1)C_{n,l+1}^+ + lC_{n,l-1}^+\right) + \Sigma_{tg}(x)\sum_{l=0}^{\infty}\phi_g^l(x)\left(\frac{2l+1}{2}\right)C_{n,l}^+\\ &= \sum_{g'=1}^{G}\sum_{l=0}^{\infty}\left(\frac{2l+1}{2}\right)\Sigma_{sgg'}^l(x)\,\phi_{g'}^l(x)C_{n,l}^+ + \frac{\chi_g}{2k_{\text{eff}}}\sum_{g'=1}^{G}\nu_g\Sigma_{fg'}(x)\phi_{g'}^0(x)C_{n,0}^+\,,\end{aligned} \tag{29}$$

where

$$C_{n,l}^+ = \int_0^{+1} P_n^+(\mu)P_l(\mu)\,d\mu\,. \tag{30}$$

Multiplying Eq. (8) by $P_n^-(\mu)$ and integrating μ between -1 and 0, we have,

$$\begin{aligned}&\sum_{l=0}^{\infty}\frac{d}{dx}\phi_g^l(x)\frac{1}{2}\left((l+1)C_{n,l+1}^- + lC_{n,l-1}^-\right) + \Sigma_{tg}(x)\sum_{l=0}^{\infty}\phi_g^l(x)\left(\frac{2l+1}{2}\right)C_{n,l}^-\\ &= \sum_{g'=1}^{G}\sum_{l=0}^{\infty}\left(\frac{2l+1}{2}\right)\Sigma_{sgg'}^l(x)\,\phi_{g'}^l(x)C_{n,l}^- + \frac{\chi_g}{2k_{\text{eff}}}\sum_{g'=1}^{G}\nu_g\Sigma_{fg'}(x)\phi_{g'}^0(x)C_{n,0}^-\,,\end{aligned} \tag{31}$$

where

$$C_{n,l}^- = \int_{-1}^{0} P_n^-(\mu)P_l(\mu)\,d\mu\,. \tag{32}$$

Using the matrix notation over energy groups defined in Eq. (12), the DP_N Eqs. (29) and (31) can be also expressed as

$$\begin{aligned}&\sum_{l=0}^{\infty}\frac{d}{dx}\phi^l\frac{1}{2}\left((l+1)C_{n,l+1}^+ + lC_{n,l-1}^+\right) + \sum_{l=0}^{\infty}\left(\frac{2l+1}{2}\right)\boldsymbol{\Sigma}^l\phi^l C_{n,l}^+\\ &\quad= \frac{1}{2k_{\text{eff}}}\mathbf{F}\phi^0 C_{n,0}^+\,,\\ &\sum_{l=0}^{\infty}\frac{d}{dx}\phi^l\frac{1}{2}\left((l+1)C_{n,l+1}^- + lC_{n,l-1}^-\right) + \sum_{l=0}^{\infty}\left(\frac{2l+1}{2}\right)\boldsymbol{\Sigma}^l\phi^l C_{n,l}^-\\ &\quad= \frac{1}{2k_{\text{eff}}}\mathbf{F}\phi^0 C_{n,0}^-\,.\end{aligned} \tag{33}$$

The sum of Eqs. (33) are truncated at $l = 2N+1$ to obtain the DP_N equations in terms of the moments ϕ^l, $l = 0,1,\ldots,2N+1$, as

$$\begin{aligned}&\sum_{l=0}^{2N+1}\frac{d}{dx}\phi^l\frac{1}{2}\left((l+1)C_{n,l+1}^+ + lC_{n,l-1}^+\right) + \sum_{l=0}^{2N+1}\left(\frac{2l+1}{2}\right)\boldsymbol{\Sigma}^l\phi^l C_{n,l}^+\\ &\quad= \frac{1}{2k_{\text{eff}}}\mathbf{F}\phi^0 C_{n,0}^+\,,\\ &\sum_{l=0}^{2N+1}\frac{d}{dx}\phi^l\frac{1}{2}\left((l+1)C_{n,l+1}^- + lC_{n,l-1}^-\right) + \sum_{l=0}^{2N+1}\left(\frac{2l+1}{2}\right)\boldsymbol{\Sigma}^l\phi^l C_{n,l}^-\\ &\quad= \frac{1}{2k_{\text{eff}}}\mathbf{F}\phi^0 C_{n,0}^-\,,\end{aligned} \tag{34}$$

where $n = 0,1,\ldots,N$. The resulting system differ from the classical ones in the considered closure since the truncation of the series is done in terms of the coefficients ϕ^l instead of $\phi^{l,+}$ and $\phi^{l,-}$ (Gelbard, 1960; Stacey, 2007).

3.2. Simplified DP_N equations

To extend the double P_N equations (Eqs. (34)) to multidimensional problems, we apply a similar procedure to the one applied for the P_N equations, that leads to the simplified double P_N equations (SDP_N equations). Therefore, the derivative with respect to x of even moments is substituted by the gradient operator, and the partial derivative of x of odd moments, by the divergence operator. The resulting set of SDP_N equations allows the odd moments of the flux to be isolated with respect to the even moments, giving a diffusive system of second-order differential equations where the even moments are the unknowns. To illustrate this process, we develop the particular case of the SDP_N equations for $N = 1$.

To obtain the SDP_1 equations, we truncate expansion (4) as

$$\psi(x,\mu) = \sum_{l=0}^{3}\left(\frac{2l+1}{2}\right)\phi^l(x)P_l(x)\,, \tag{35}$$

and the Eqs. (34) are considered for $n = 0,1$ (DP_1 equations). The simplified formulation is developed by changing the derivative of x by gradient or divergence operator. The SDP_1 equations are

$$\begin{aligned}&\frac{1}{4}\nabla\phi^0 + \frac{1}{2}\nabla\cdot\phi^1 + \frac{5}{16}\nabla\phi^2\frac{1}{2}\boldsymbol{\Sigma}^{\mathbf{0}}\phi^0 + \frac{3}{4}\boldsymbol{\Sigma}^{\mathbf{1}}\phi^1 - \frac{7}{16}\boldsymbol{\Sigma}^{\mathbf{3}}\phi^3 = \frac{1}{2k_{\text{eff}}}\mathbf{F}\phi^0,\\ &-\frac{1}{4}\nabla\phi^0 + \frac{1}{2}\nabla\cdot\phi^1 - \frac{5}{16}\nabla\phi^2 + \frac{1}{2}\boldsymbol{\Sigma}^{\mathbf{0}}\phi^0 - \frac{3}{4}\boldsymbol{\Sigma}^{\mathbf{1}}\phi^1 + \frac{7}{16}\boldsymbol{\Sigma}^{\mathbf{3}} = \frac{1}{2k_{\text{eff}}}\mathbf{F}\phi^0,\\ &\frac{1}{12}\nabla\phi^0 + \frac{1}{4}\nabla\cdot\phi^1 + \frac{17}{48}\nabla\phi^2 + \frac{7}{24}\nabla\cdot\phi^3 + \frac{1}{4}\boldsymbol{\Sigma}^{\mathbf{1}}\phi^1 + \frac{5}{8}\boldsymbol{\Sigma}^{\mathbf{2}}\phi^2\\ &\qquad+\frac{7}{16}\boldsymbol{\Sigma}^{\mathbf{3}}\phi^3 = 0\,,\\ &\frac{1}{12}\nabla\phi^0 - \frac{1}{4}\nabla\cdot\phi^1 + \frac{17}{48}\nabla\phi^2 - \frac{7}{24}\nabla\cdot\phi^3 + \frac{1}{4}\boldsymbol{\Sigma}^{\mathbf{1}}\phi^1 - \frac{5}{8}\boldsymbol{\Sigma}^{\mathbf{2}}\phi^2\\ &\qquad+\frac{7}{16}\boldsymbol{\Sigma}^{\mathbf{3}}\phi^3 = 0\,.\end{aligned} \tag{36}$$

From Eqs. (36), the odd moments of the neutron flux can be isolated in terms of even moments as

$$\phi^1 = -\frac{1}{3}\left(\boldsymbol{\Sigma}^{\mathbf{1}}\right)^{-1}\left(\nabla\phi^0 + 2\nabla\phi^2\right), \tag{37}$$

$$\phi^3 = -\frac{3}{7}\left(\boldsymbol{\Sigma}^{\mathbf{3}}\right)^{-1}\nabla\phi^2, \tag{38}$$

and, these expressions can then be substituted into the equations of the even moments reducing the size of the initial system (36) by half. The resulting problem of differential equations is

$$-\frac{1}{3}\nabla\cdot\left(\boldsymbol{\Sigma}^{\mathbf{1}}\right)^{-1}\left(\nabla\phi^0 + 2\nabla\phi^2\right) + \Sigma^0\phi^0 = \frac{1}{k_{\text{eff}}}\mathbf{F}\phi^0, \tag{39}$$

$$-\frac{2}{15}\nabla\cdot\left(\boldsymbol{\Sigma}^{\mathbf{1}}\right)^{-1}\left(\nabla\phi^0 + 2\nabla\phi^2\right) - \frac{1}{5}\nabla\cdot\left(\boldsymbol{\Sigma}^{\mathbf{3}}\right)^{-1}\nabla\phi^2 + \Sigma^2\phi^2 = 0. \tag{40}$$

This system can also be transformed into a system of diffusion-like equations by introducing the same change of variables proposed for the SP_N Eqs. (13) for $M = 2$, to obtain

$$-\nabla\cdot(\mathbb{D}\nabla U) + \mathbb{A}U = \frac{1}{k_{\text{eff}}}\mathbb{F}U\,, \tag{41}$$

where

$$\mathbb{D} = \begin{pmatrix}\frac{1}{3}(\boldsymbol{\Sigma}^{\mathbf{1}})^{-1} & 0\\ 0 & \frac{1}{7}(\boldsymbol{\Sigma}^{\mathbf{3}})^{-1}\end{pmatrix},\quad \mathbb{A}_{ij} = \sum_{m=1}^{2}\mathbf{c}_{ij}^{(\mathbf{m})}\boldsymbol{\Sigma}^{\mathbf{m}},\quad \mathbb{F}_{ij} = \mathbf{c}_{ij}^{(\mathbf{1})}\mathbf{F}, \tag{42}$$

and the coefficients

$$\mathbf{c}^{(\mathbf{1})} = \begin{pmatrix}1 & -\frac{2}{3}\\ -\frac{6}{7} & \frac{4}{7}\end{pmatrix},\quad \mathbf{c}^{(\mathbf{2})} = \begin{pmatrix}0 & 0\\ 0 & \frac{5}{7}\end{pmatrix}. \tag{43}$$

The expressions for SDP_2 and SDP_3 equations can be found in Appendix. These expressions are different from the ones presented in Nazari et al. (2023) because a different closure is considered, as it has been mentioned above.

3.3. Boundary conditions for SDP_N

Vacuum boundary conditions for DP_N equations are also extended by applying Marshak conditions, but considering the polynomials (27). They are defined such that the incoming flux moments are zero at the boundary. In this way, for the left boundary condition at $x = x_L$, we have

$$\int_0^1 P_n^+(\mu)\psi_g^{\text{in}}(x_L,\mu)\,d\mu = \int_0^1 P_n^+(\mu)\sum_{l=0}^{\infty}\phi_g^l\left(x_L\right)\left(\frac{2l+1}{2}\right)P_l(\mu)\,d\mu$$

$$= \sum_{l=0}^{\infty}\left(\frac{2l+1}{2}\right)\phi_g^l\left(x_L\right)C_{n,l}^+ = 0, \qquad g = 1,\ldots,G, \quad n = 0,1,\ldots,N. \tag{44}$$

A similar condition is obtained for the right boundary condition in $x = x_R$.

For DP_1 equations, the vacuum boundary conditions (44) are

$$\frac{1}{2}\phi^0 + \frac{5}{8}\phi^2 = -\phi^1,$$
$$-\frac{4}{14}\phi^0 + \frac{15}{14}\phi^2 = -\phi^3. \tag{45}$$

Applying the same procedure as for the boundary conditions of SP_N equations (Section 2.2), they are also expressed as

$$-\vec{n}\,\mathbb{D}\vec{\nabla}U = \mathbb{B}U. \tag{46}$$

where the matrix $\mathbb{B}$ is, in this case of $N = 1$, given by

$$\mathbb{B} = \mathbf{g}\otimes\mathbf{I}_{(G\times G)}, \qquad \text{where} \qquad \mathbf{g} = \begin{pmatrix} \frac{1}{2} & -\frac{1}{8} \\ -\frac{2}{7} & \frac{23}{42} \end{pmatrix}. \tag{47}$$

Reflective boundary conditions for the 1D DP_N equations are applied by imposing that $\phi^n(x_L) = 0 = \phi^n(x_R)$, $n = 1,3,\ldots$ as in the P_N equations (Section 2.2). Therefore, these boundary conditions for the SDP_N equations are also imposed as

$$\nabla U^m = 0, \qquad m = 1,2,\ldots,(N+1)/2. \tag{48}$$

Boundary conditions for SDP_2 and SDP_3 equations can be found in Appendix.

4. Numerical results

To assess the accuracy and the performance of the SDP_N equations, these equations are implemented in the neutronic code Finite Element Method `FEMFFUSION` (Vidal-Ferràndiz et al., 2020) by using structures of the open-source `deal.ii` (Bangerth et al., 2007) and `PETSc` (Balay et al., 2015) libraries. In particular, a continuous Galerkin finite element method with Lagrange polynomials is used to obtain an algebraic eigenvalue problem. Moreover, a matrix-free strategy is applied in the implementation that avoids full storage of the matrices by computing matrix–vector products on the fly (Vidal-Ferràndiz et al., 2019). The resulting algebraic eigensystem is solved with the BIFPAM algorithm and the block Gauss–Seidel preconditioner (Vidal-Ferràndiz et al., 2019).

Three different benchmark problems are used to analyze the approximations of the neutron transport equation studied. In order to compare the solutions in these problems, three types of errors are used: the k-effective error (Δk_{eff}), the average cell error (AVG) and the root-mean-square error (RMSE). They are computed as

$$\Delta k_{\text{eff}} = \frac{|k_{\text{eff}}^{\text{ref}} - k_{\text{eff}}|}{k_{\text{eff}}^{\text{ref}}}\cdot 10^5, \tag{49}$$

$$\text{AVG} = \frac{1}{N_{cells}}\sum_{i=1}^{N_{cells}}\left|\frac{P_i^{\text{ref}} - P_i}{P_i^{\text{ref}}}\right|, \tag{50}$$

$$\text{RMSE} = \frac{1}{N_{cells}}\sum_{i=1}^{N_{cells}}\sqrt{\left(\frac{P_i^{\text{ref}} - P_i}{P_i^{\text{ref}}}\right)^2}, \tag{51}$$

where N_{cells} is the total number of fuel cells where $P_i^{\text{ref}} \neq 0$ and P_i denotes the average neutron flux or power in the ith cell.

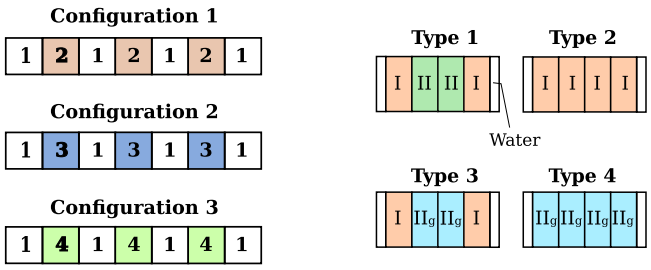


Fig. 1. Configurations of the 1D BWR problems (Configuration 1, 2 and 3).

Table 1
Material properties for the 1D BWR benchmark.
Source: Sourced from Rahnema and Nichita (1997).

Material	Thickness (cm)	g	D_g (cm)	Σ_{ag} (1/cm)	$\nu\Sigma_{fg}$ (1/cm)	Σ_{12} (1/cm)
Water	1.158	1	1.7639	0.0003	0.0000	0.0380
		2	0.2278	0.0097	0.0000	
Fuel I	3.231	1	1.4730	0.0096	0.0067	0.0161
		2	0.3294	0.0764	0.1241	
Fuel II	3.231	1	1.4804	0.0101	0.0078	0.0156
		2	0.3362	0.0901	0.1542	
Fuel IIg	3.231	1	1.5342	0.0135	0.0056	0.0136
		2	0.3143	0.4873	0.0187	

4.1. 1D BWR problems

First, one-dimensional Boiling Water Reactor (BWR) benchmark problems composed of 7 assemblies with heterogeneous regions extracted from Rahnema and Nichita (1997) are studied. The reactor is built with three distinct configurations, formed by combining three different assemblies to model a BWR lattice. Their geometry is shown in Fig. 1. Cross-sectional data for each assembly region can be found in Table 1. The configurations are organized on the basis of the heterogeneity among assemblies. Configuration 1 consists of similar assemblies, while Configurations 2 and 3 join highly heterogeneous components. Assemblies of types 3 and 4, present in Configurations 2 and 3, incorporate gadolinium in some or all of the fuel regions, given its high neutron absorption properties. It is important to highlight that assembly type 4 deviates from the usual BWR assembly. Its inclusion in Configuration 3 aims to establish a highly heterogeneous core geometry with significant neutronic flux gradients, serving as a limiting condition for low-order angular discretizations. Zero-current boundary conditions are enforced at the system edges.

Reference transport results are extracted using the Method of Characteristics code, `OpenMOC` (Boyd et al., 2014). As `OpenMOC` does not support one-dimensional calculations, an analogous two-dimensional mesh with symmetry boundary conditions is used. A fine mesh of 1050×25 cells with 32 azimuthal angles and a track spacing of 0.0125 cm is used to obtain accurate reference results. To spatially discretize the SP_N and SDP_N equations, finite element results are computed using a coarser grid of 42×1 cells and third-degree polynomials. This grid ensures spatially converged results for this type of numerical benchmarks (Vidal-Ferràndiz et al., 2019).

Numerical results for the Configuration 1 of the 1D BWR benchmark are shown in Table 2 using the SDP_N equations, up to SDP_3 and the SP_N equations up to SP_7, including also the size of the problem (Number of Degrees of Freedom (N Dofs)). Note that SDP_1 and SP_3 result in an eigenvalue problem of the same size. Analogously, SDP_2 and SP_5 approximations; and SDP_3 and SP_7 produce matrices of the same dimension. It can be seen that as the spherical harmonic order, N increases, the results converge to the reference transport results the SP_N as well as the SDP_N equations. Approximations with the same number of DoFs produce similar results, but the SDP_N equations provide slightly

Table 2
Numerical results for the one dimensional problem Configuration 1.

Approx.	N DoFs	k-effective	Δk_{eff} (pcm)	RMSE g_1 (%)	RMSE g_2 (%)	CPU time (s)
SDP_1	508	1.317 54	68	0.98	0.67	0.06
SDP_2	762	1.317 11	25	0.37	0.55	0.13
SDP_3	1016	1.316 92	6	0.13	0.53	0.20
SP_1	254	1.319 92	306	2.8	0.91	0.03
SP_3	508	1.317 75	89	1.24	0.72	0.06
SP_5	762	1.317 31	45	0.64	0.59	0.12
SP_7	1016	1.317 11	25	0.36	0.55	0.22
OpenMOC		1.316 86				

Table 3
Numerical results for the one dimensional problem Configuration 2.

Approx.	N DoFs	k-effective	Δk_{eff} (pcm)	RMSE g_1 (%)	RMSE g_2 (%)	CPU time (s)
SDP_1	508	1.124 84	54	1.07	1.08	0.16
SDP_2	762	1.124 43	13	0.44	0.99	0.23
SDP_3	1016	1.124 23	7	0.22	0.82	0.83
SP_1	254	1.120 78	352	3.3	2.46	0.05
SP_3	508	1.124 85	55	1.3	1.1	0.14
SP_5	762	1.124 59	29	0.71	1.02	0.23
SP_7	1016	1.124 42	12	0.43	0.99	0.69
OpenMOC		1.124 30				

Table 4
Numerical results for the one dimensional problem Configuration 3.

Approx.	N DoFs	k-effective	Δk_{eff} (pcm)	RMSE g_1 (%)	RMSE g_2 (%)	CPU time (s)
SDP_1	508	1.029 92	50	0.89	2.22	0.14
SDP_2	762	1.029 58	16	0.46	1.23	1.06
SDP_3	1016	1.029 41	1	0.35	0.24	1.39
SP_1	254	1.0218	762	5.23	5.89	0.05
SP_3	508	1.029 79	37	1.16	2.24	0.27
SP_5	762	1.029 69	27	0.63	1.22	0.34
SP_7	1016	1.029 56	14	0.45	0.57	2.27
OpenMOC		1.029 42				

better results, mainly for the fast neutron flux ($g = 1$). For example, SDP_3 results provide 6 pcm of error and RSMEs of 0.13% and 0.53% for $g = 1$ and $g = 2$, while the SP_7 gives 25 pcm of error in the eigenvalue calculations and RMSEs of 0.36% and 0.55% by using almost identical CPU times.

Fig. 2 shows the fast neutron flux for the Configuration 1 of the 1D BWR problem. We compare the approximation of the same N Dofs in the same graphic with the `OpenMOC` transport reference solution. These Figures also show the convergence of both formulations. The errors can be found mainly in the relative minimums of the scalar neutron flux produced by the water gaps between assemblies where a high heterogeneity is present (high flux gradients).

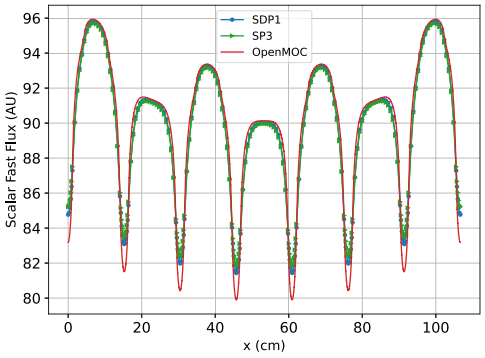
Tables 3 and 4 provide the results for Configurations 2 and 3, respectively. Similar conclusions as in Configuration 1 are obtained, but as these problems get more heterogeneous, the improvement of SDP_N equations are more significant.

4.2. Two-dimensional one-group problem

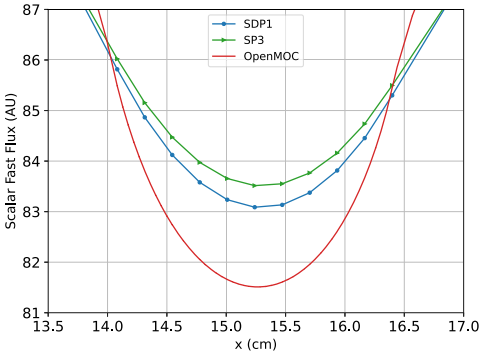
Secondly, a one-group, isotropic-scattering eigenvalue two-dimensional problem is considered (Brantley and Larsen, 2000). In this problem, diffusive approximations are not strictly valid. Fig. 3 shows the geometry of this nuclear system. Table 5 contains the monoenergetic cross-section data used. Due to the geometry and the reflective conditions of the problem, it has a one-dimensional behavior along the line $y = 0$ and progressively shows two-dimensional nature as y increases. The SDP_N, with $N = 1, 2, 3$, and SP_N, with $N = 1, 3, 5, 7$, equations are solved. For the spatial discretization, a uniform mesh of 80×80 cells of dimension 0.125×0.125 cm^2 is considered. Polynomials of degree two are used in the finite element method results. Reference values are obtained by using the method of the characteristics obtained with OpenMOC code (Boyd et al., 2014) with a fine mesh of 250×250 cells, 32 azimuthal angles and a track spacing of 0.0125 cm.

Table 6 shows the computed criticality values, k_{eff}, the eigenvalue errors and the scalar flux errors. Moreover, it is included for comparison the discrete ordinates k-effective from the `TWODANT` code (Alabama A & M University et al., 1995) with a S_{16} quadrature and a 150×150 spatial mesh (Brantley and Larsen, 2000). The solution obtained with the simplified double P_N formulation proposed in Nazari et al. (2023) is also added. Numerical results show that errors are reduced as N increases. However, the improvement stagnates around a RMSE = 0.75% in SDP_2, SDP_3, SP_5 and SP_7. This effect is expected because SPN is an asymptotic approximation, which implies that, in general, there is a N where the smallest error is obtained. If approximations of the same number of DoFs are compared, it exists an improvement between SP_3 and SDP_1, SP_5 and SDP_2, and SP_7 and SDP_3 in terms on the k-effective, even though these differences cannot be observed in the neutron flux. Similar results, in terms of the error and CPU times, are obtained in the comparison with formulation developed in Nazari et al. (2023).

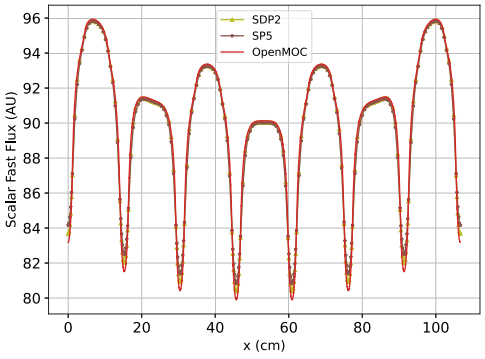
Fig. 4 represents the scalar neutron flux along $y = 4.5$ cm for the one-group eigenvalue problem for SDP_3, SP_3 (Nazari et al., 2023), SP_7 approximations, and the method of characteristics (`OpenMOC`). This Figure does not show relevant differences between the approximations.



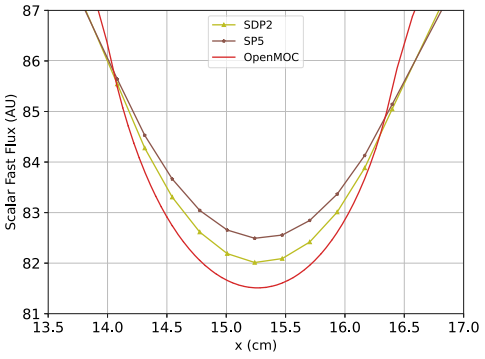
(a) SPD1 and SP_3 comparison for $g = 1$



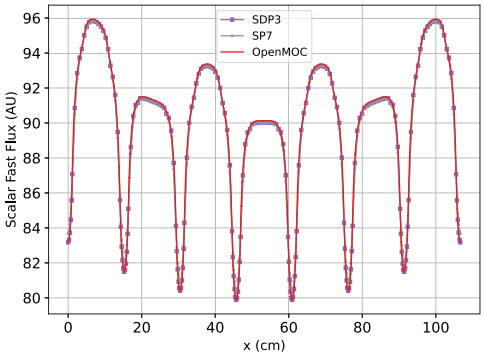
(b) SPD1 and SP_3 comparison for $g = 1$ where $x \in [13.5, 17]$ cm



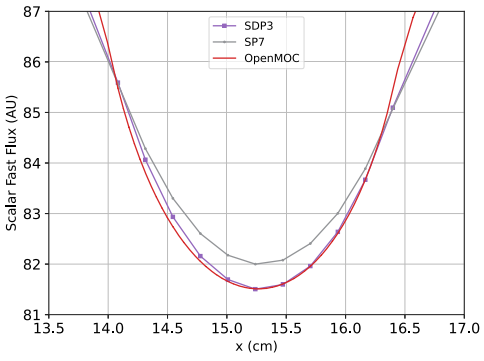
(c) SPD2 and SP_5 comparison for $g = 1$



(d) SPD2 and SP_5 comparison for $g = 1$ where $x \in [13.5, 17]$ cm



(e) SPD3 and SP_7 comparison for $g = 1$



(f) SPD3 and SP_7 comparison for $g = 1$ where $x \in [13.5, 17]$ cm

Fig. 2. Scalar flux results in the Configuration 1 of 1D BWR problems.

Table 5
Nuclear data one-group problem.
Source: Sourced from Brantley and Larsen (2000).

Data	Moderator	Fuel
Σ_t (cm^{-1})	1.00	1.500
Σ_{s0} (cm^{-1})	0.93	1.350
Σ_f (cm^{-1})	0.00	0.100
$\nu\Sigma_f$ (cm^{-1})	0.00	0.240

4.3. Two-dimensional C5G7

We conclude this analysis by using the two-dimensional C5G7 benchmark defined by the Nuclear Energy Agency (NEA) (Nuclear Energy Agency (NEA), 2003). This problem represents a realistic lattice of 4 fuel assemblies. The solution to this benchmark has been approximated by several types of neutron transport codes, and the solution obtained by the Monte Carlo method is used as reference.

This nuclear reactor core is configured with two fuel types, MOX and UO_2, which are surrounded by a moderated region (Fig. 5). The material data of this reactor can be found in Nuclear Energy Agency (NEA) (2003). Each square fuel assembly contains 17×17 pin structures with

Table 6
Numerical results for one-group problem.

Approximation	k-effective	Δk_{eff} (pcm)	RMSE (%)	CPU time (s)
SDP_1	0.801 61	375	0.92	1.5
SDP_2	0.803 73	163	0.76	2.0
SDP_3	0.804 02	134	0.76	2.6
SP_1	0.776 80	2856	4.16	1.1
SP_3	0.799 04	632	1.01	1.5
SP_5	0.802 80	256	0.74	2.0
SP_7	0.803 54	182	0.74	2.6
SDP_1 (Nazari et al., 2023)	0.806 85	149	0.84	1.5
SDP_2 (Nazari et al., 2023)	0.804 08	128	0.78	2.1
SDP_3 (Nazari et al., 2023)	0.804 11	125	1.21	13.2
OpenMOC	0.805 36			
S_{16} (Brantley and Larsen, 2000)	0.806 13			

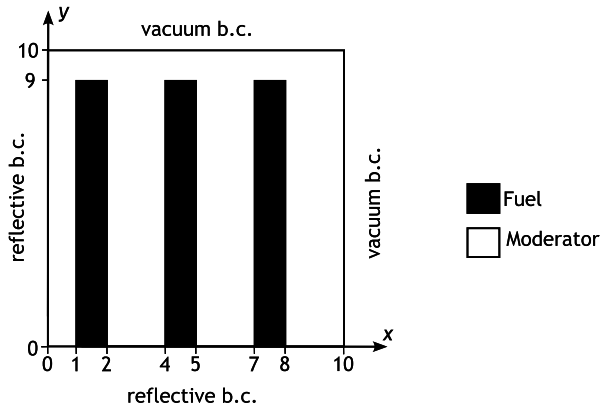


Fig. 3. Geometry of the two-dimensional one-group problem.

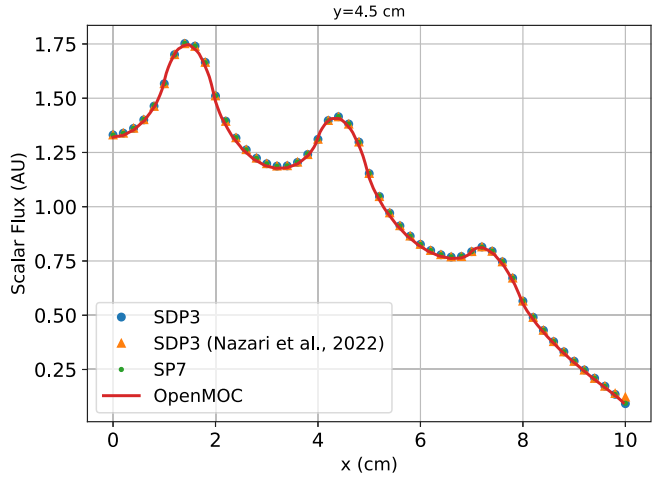


Fig. 4. Scalar flux along $y = 4.5$ cm obtained with SDP_3, SP_3 (Nazari et al., 2023), SP_7 and OpenMOC for the two-dimensional one-group problem.

dimensions of 1.26×1.26 cm^2. Each pin consists of a circular fuel area with a radius of 0.54 cm, surrounded by a moderator region. The geometry and the spatial discretization of each pin are also shown in Fig. 5. Second degree polynomials are used in the finite element calculations to ensure that the numerical results are spatially converged (Fontenla et al., 2024; Vidal-Ferràndiz et al., 2019).

Table 7 shows the numerical results of using the approximation SDP_N with $N = 1, 2, 3$ and the approximation SP_N with $N = 1, 3, 5, 7$. It can be seen that as N increases, the results do not improve as they do in the previous benchmarks. This behavior has also been observed in previous works (Vidal-Ferràndiz et al., 2019; Carreño et al., 2022) where the SP_5 and the SP_7 approximations do not improve the precision of the solution with respect to SP_3. Same results are observed in the Simplified Double P_N approximations, where SDP_2 and SDP_3 do not improve significantly the results of SDP_1.

Figs. 6 and 7 compare the scalar fluxes for $g = 1$ and $g = 7$ of SDP_N approximation along the main diagonal of (1, 1) UO_2 fuel assembly and along the antidiagonal of (2, 1) MOX fuel assembly, respectively. Moreover, these graphics include a reference solution for the neutronic scalar flux obtained with the code OpenMC (Romano et al., 2015). These Figures show that higher SDP_N approximations represent more accurately the local extremes of the neutron flux.

5. Conclusions

This work develops a simplified formulation of the double spherical harmonics equation, which is based on the expansion of angular fluxes introducing two sets of Legendre polynomials, in order to improve the approximations obtained with the classical simplified spherical harmonics equations without increasing the CPU demands, especially for reactors composed of heterogeneous media. This approximation leads to a system of diffusion-type differential equations of the second order, the simplified double spherical harmonics equations. A new formulation of the SDP_N equations is introduced leading to a set of diffusive equations where the different order approximations can be formulated in a unique matrix structure as it happens with the SP_N equations. This is convenient for the development of multilevel methods based on different angular approximations.

Numerical results show the convergence of the SDP_N approximation to the transport solution as N increases. In general, the simplified double P_N equations provide more accurate results than SP_N equations if approximations with the same number of equations are compared. Therefore, SDP_1 equations provide better results than SP_3 equations; SDP_2 give more precise results than SP_5; and SDP_3 improve the SP_7 results. Moreover, the CPU times are reduced in the previous comparisons because the obtained eigenvalue problems reach the convergence in less number of iterations. Nevertheless, in instances where the simplified approximations exhibit stagnation in the convergence, the Simplified Double P_N equations do not provide a great improvement with respect to SP_N approximation. This is the case of the C5G7 benchmark problem.

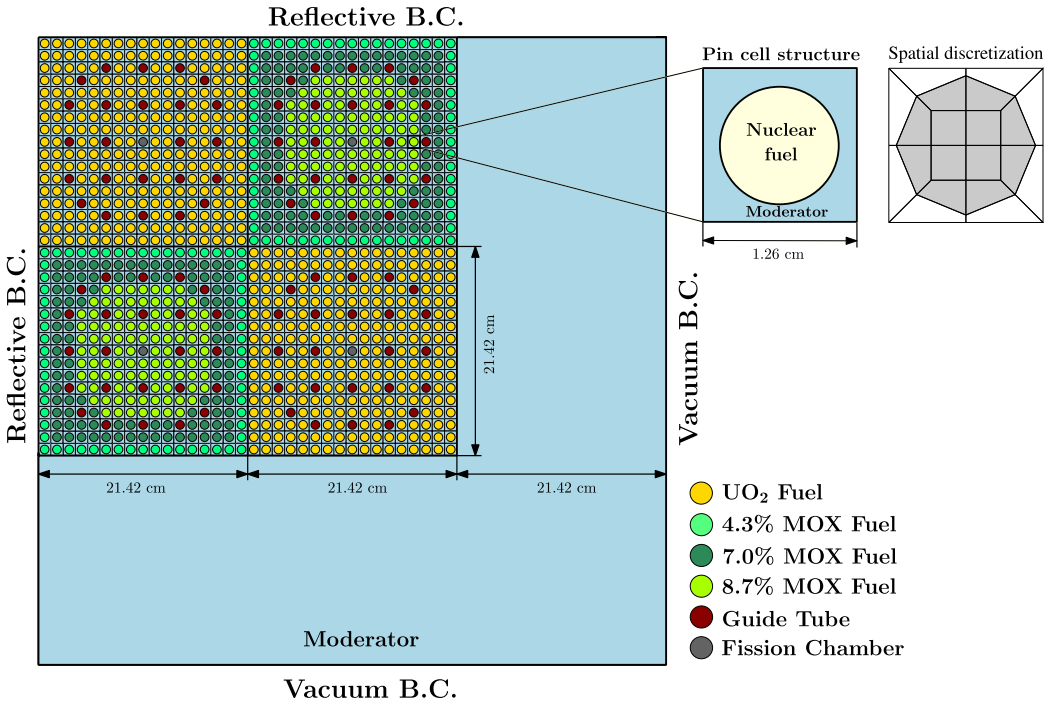
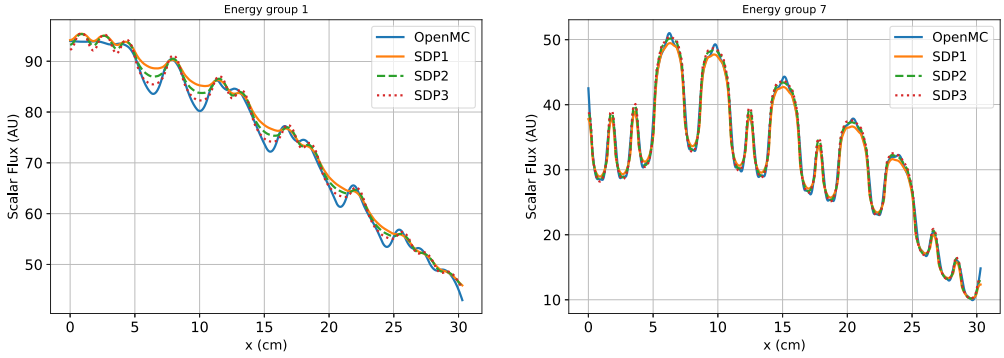
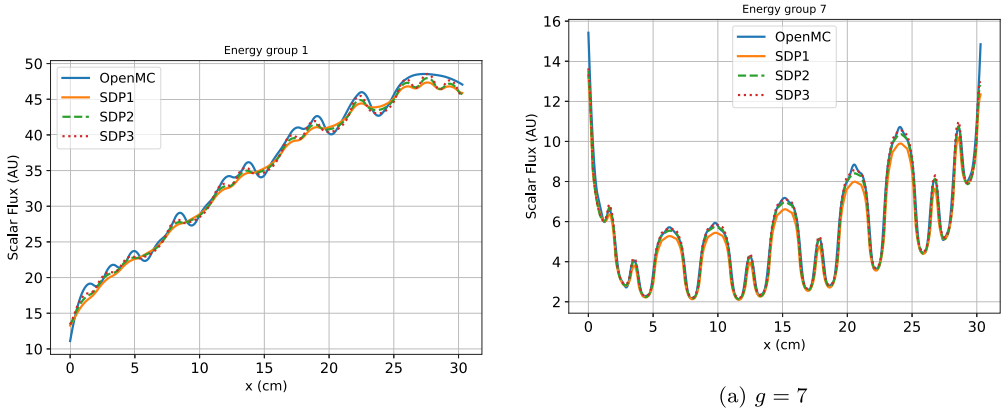


Fig. 5. Geometry of the C5G7 benchmark.



(a) $g = 1$ (b) $g = 7$

Fig. 6. Scalar neutron flux along the main diagonal of (1, 1) UO_2 fuel assembly.



(a) $g = 7$

Fig. 7. Scalar neutron flux along the anti-diagonal of (2, 1) MOX fuel assembly.

Table 7
Numerical results for C5G7 problem.

Approximation	k-effective	Δk_{eff} (pcm)	AVG (%)	RMSE (%)
SDP_1	1.182 42	413	0.68	0.03
SDP_2	1.182 19	436	0.57	0.02
SDP_3	1.182 86	369	0.64	0.03
SP_1	1.183 36	319	1.43	0.05
SP_3	1.182 71	384	0.75	0.03
SP_5	1.182 21	434	0.59	0.02
SP_7	1.182 31	424	0.57	0.02
OpenMC	1.18655			
MCNP Nuclear Energy Agency (NEA) (2003)	1.186 55			

CRediT authorship contribution statement

A. Carreño: Writing – original draft, Software, Methodology, Formal analysis. **A. Vidal-Ferràndiz:** Writing – original draft, Software, Methodology, Investigation. **D. Ginestar:** Writing – review & editing, Methodology, Formal analysis. **G. Verdú:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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Appendix. Solution of the SDP_2 and SDP_3 equations

The SDP_2 equations are obtained by considering the expansion of the angular neutron flux as

$$\psi(x,\mu)=\sum_{l=0}^{5}\phi^l(x)\left(\frac{2l+1}{2}\right)P_l(x). \tag{A.1}$$

If a similar process is applied as for the SDP_1 equations and the transformation of variables (13) is considered, one can get a system of the form (41) where, in this case,

$$U=\left(U^1,U^2,U^3\right)^{\top},\quad \mathbb{A}_{ij}=\sum_{m=1}^{3}\mathbf{c}_{ij}^{(\mathbf{m})}\boldsymbol{\Sigma}^{\mathbf{m}},\quad \mathbb{F}_{ij}=\mathbf{c}_{ij}^{(\mathbf{1})}\mathbf{F},$$
$$\mathbb{D}=\begin{pmatrix}\frac{1}{3}(\boldsymbol{\Sigma}^{\mathbf{1}})^{-1} & 0 & 0\\ 0 & \frac{1}{7}(\boldsymbol{\Sigma}^{\mathbf{3}})^{-1} & 0\\ 0 & 0 & \frac{1}{11}(\boldsymbol{\Sigma}^{\mathbf{5}})^{-1}\end{pmatrix}, \tag{A.2}$$

and the coefficient matrices are

$$\mathbf{c}^{(\mathbf{1})}=\begin{pmatrix}1 & -\frac{2}{3} & \frac{8}{15}\\ -\frac{2}{3} & \frac{4}{9} & -\frac{16}{45}\\ -\frac{32}{33} & -\frac{64}{99} & \frac{256}{495}\end{pmatrix},\quad \mathbf{c}^{(\mathbf{2})}=\begin{pmatrix}0 & 0 & 0\\ 0 & \frac{5}{9} & -\frac{4}{9}\\ 0 & -\frac{80}{99} & \frac{64}{99}\end{pmatrix},$$
$$\mathbf{c}^{(\mathbf{3})}=\begin{pmatrix}0 & 0 & 0\\ 0 & 0 & 0\\ 0 & 0 & \frac{36}{55}\end{pmatrix}. \tag{A.3}$$

In the SDP_2, the vacuum boundary conditions can be applied through the use of the Eqs. (46) and (47), but the matrix $\mathbf{g}$ is new given by

$$\mathbf{g}=\begin{pmatrix}\frac{1}{2} & -\frac{1}{8} & \frac{1}{16}\\ -\frac{2}{21} & \frac{31}{126} & -\frac{59}{1260}\\ \frac{8}{33} & -\frac{38}{99} & \frac{287}{495}\end{pmatrix}. \tag{A.4}$$

Reflective conditions are imposed by setting Eq. (48).

Similarly, one can obtain that the matrices of the system (41) for the SDP_3 approximation are

$$U=\left(U^1,U^2,U^3,U^4\right)^{\top},\quad \mathbb{A}_{ij}=\sum_{m=1}^{4}\mathbf{c}_{ij}^{(\mathbf{m})}\boldsymbol{\Sigma}^{\mathbf{m}},\quad \mathbb{F}_{ij}=\mathbf{c}_{ij}^{(\mathbf{1})}\mathbf{F},$$
$$\mathbb{D}=\begin{pmatrix}\frac{1}{3}(\boldsymbol{\Sigma}^{\mathbf{1}})^{-1} & 0 & 0 & 0\\ 0 & \frac{1}{7}(\boldsymbol{\Sigma}^{\mathbf{3}})^{-1} & 0 & 0\\ 0 & 0 & \frac{1}{11}(\boldsymbol{\Sigma}^{\mathbf{5}})^{-1} & 0\\ 0 & 0 & 0 & \frac{1}{15}(\boldsymbol{\Sigma}^{\mathbf{7}})^{-1}\end{pmatrix}, \tag{A.5}$$

where the matrices $\mathbf{c}^{(\mathbf{m})}$ are given by

$$\mathbf{c}^{(\mathbf{1})}=\begin{pmatrix}1 & -\frac{2}{3} & \frac{8}{15} & -\frac{16}{35}\\ -\frac{2}{3} & \frac{4}{9} & -\frac{16}{45} & \frac{32}{105}\\ \frac{8}{15} & -\frac{16}{45} & \frac{64}{225} & -\frac{128}{525}\\ -\frac{176}{125} & \frac{352}{375} & -\frac{1408}{1875} & \frac{2816}{4375}\end{pmatrix},\quad \mathbf{c}^{(\mathbf{2})}=\begin{pmatrix}0 & 0 & 0 & 0\\ 0 & \frac{5}{9} & -\frac{4}{9} & \frac{8}{21}\\ 0 & -\frac{4}{9} & \frac{16}{45} & -\frac{32}{105}\\ 0 & \frac{88}{75} & -\frac{352}{375} & \frac{704}{875}\end{pmatrix},$$
$$\mathbf{c}^{(\mathbf{3})}=\begin{pmatrix}0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0\\ 0 & 0 & \frac{36}{55} & \frac{54}{175}\\ 0 & 0 & -\frac{594}{625} & \frac{3564}{4375}\end{pmatrix},\quad \mathbf{c}^{(\mathbf{4})}=\begin{pmatrix}0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0\\ 0 & 0 & 0 & \frac{143}{175}\end{pmatrix}. \tag{A.6}$$

In the SDP_3 equations, vacuum b.c. are applied with the condition (46) where

$$\mathbf{g}=\begin{pmatrix}\frac{1}{2} & -\frac{1}{8} & \frac{1}{16} & -\frac{5}{128}\\ -\frac{1}{8} & \frac{7}{24} & -\frac{41}{384} & \frac{1}{16}\\ -\frac{2}{143} & \frac{4}{429} & \frac{1087}{17160} & \frac{3611}{40040}\\ -\frac{10}{39} & \frac{46}{117} & -\frac{2417}{4680} & \frac{1891}{2730}\end{pmatrix}. \tag{A.7}$$

Reflecting boundary conditions are applied by using Eq. (48).

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